1.1 INTRODUCTION:

All living organisms must need to exchange oxygen and carbon dioxide gases with their environment to carry out their vital functions such as respiration. In addition to respiration, photoautotrophs like plants do carry out gaseous exchange for the process of photosynthesis. Aquatic organisms exchange gases with water while terrestrial ones with air.

1.2 GASEOUS EXCHANGE IN PLANTS

As stated above, plants exchange gases for the processes of photosynthesis and respiration.

During the process of photosynthesis, carbon dioxide is taken in while oxygen is given out whereas in respiration, oxygen is taken in and carbon dioxide is given out. During daytime, green parts of the plants carry out the process of Photosynthesis to prepare complex food molecules (organic molecules) by utilizing simple molecules such as carbon dioxide gas and water. During this process, carbon dioxide gas is taken in while oxygen gas released as by-product is given out. Respiration on the other hand, takes place in all living cells. It is the process in which food is oxidized to release energy. In aerobic respiration, it involves taking in of oxygen and given out of carbon dioxide. The process of exchange of gases in plants takes place mainly through minute openings called stomata present in leaves. The roots and stem do exchange gases for respiration.

Photosynthesis	Respiration
1. Anabolic process.	1. Catabolic process.
2. Synthesis of food from simple, inorganic substances.	2. Breaking down of food into inorganic substances.
3. Requires light energy.	3. Does not require light energy.
4. Occurs in plants.	4. Occurs in all living organisms.
5. Uses carbon dioxide gas.	5. Uses oxygen gas.
6. Releases oxygen gas.	6. Releases carbon dioxide gas.
7. Takes place during day-time.	7. Takes place all the times.
8. Chlorophyll is required.	8. Chlorophyll not required.

Stomata: (Singular Stoma means mouth)

These are microscopic openings present in the epidermis of leaves. Through these openings, plants exchange gases with their environment. Each stoma is a slit like opening formed by two special cells called **guard cells**. They are chlorophyll containing cells with thicker inner while thinner and elastic outer cell walls. The opening and closing of stomata depends upon turgidity of the guard cells. During daytime, as a result of ongoing process of photosynthesis, the accumulation of photosynthetic solutes causes increase in turgidity of the guard cells. Thus stomata are opened and the process of taking in of carbon dioxide and giving out of oxygen begins until it becomes dark.

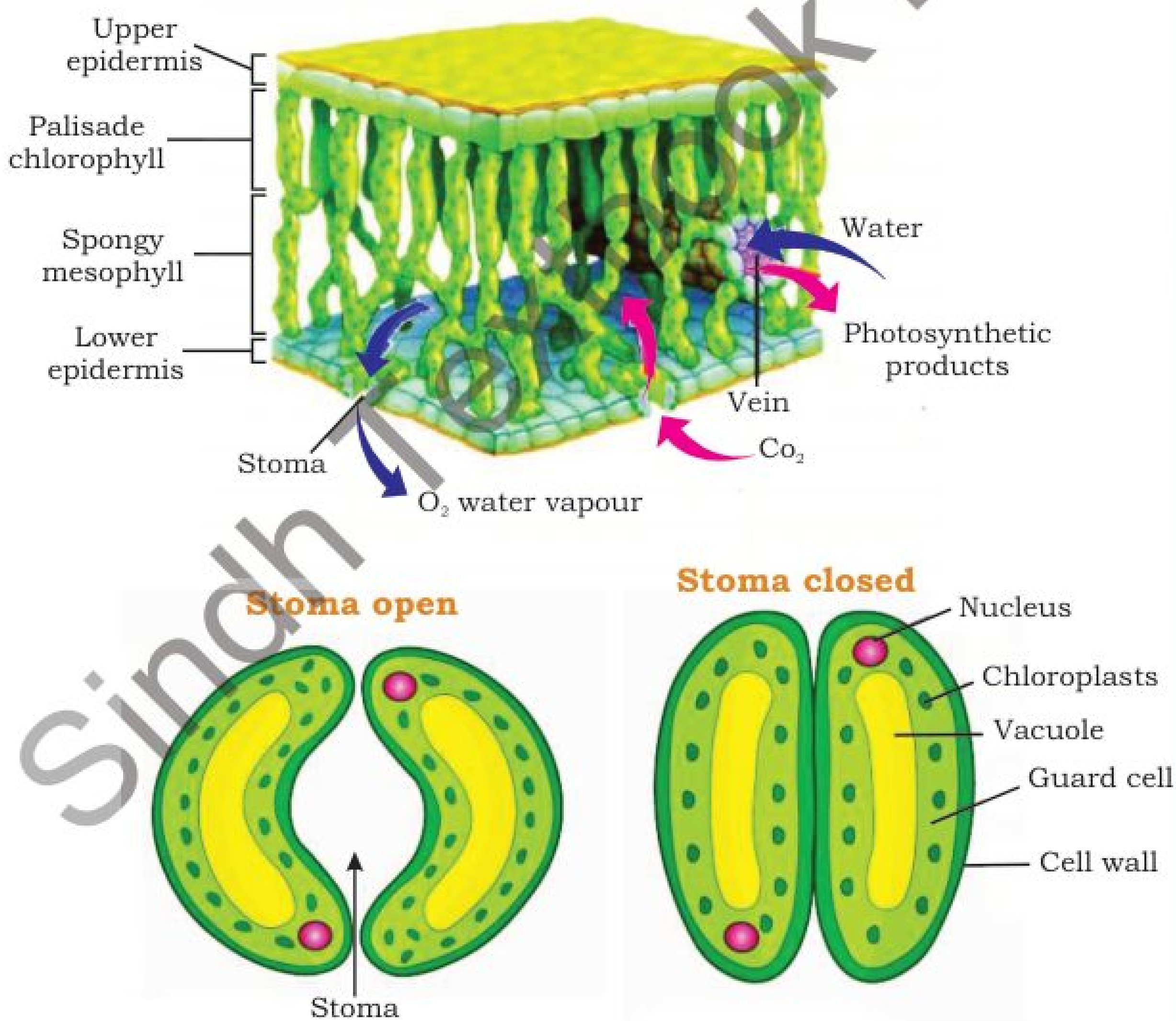


Fig. 1.1 Gaseous Exchange

PRACTICAL ACTIVITY

The effect of light on the net gaseous exchange from leaf by using Hydrogen bicarbonate as indicator.

Hydrogen bicarbonate is an indicator for carbon dioxide. Its colour turns as follows according to the level of carbon dioxide:

Photosynthesis	Respiration
Highest	Yellow
High	Orange
Atmospheric level	Red
Low	Magenta
Lowest	Purple

Requirements:

Four test tubes, test tube stand, aluminum foils or black paper, tissue paper, fresh green leaves, four corks, wax, thread, glass marking pencil

Steps:

1. Mark test tubes as 1, 2, 3 and 4.
2. Fill each test tube quarter full with Hydrogen bicarbonate indicator.
3. Attach each leaf with separate thread and hang one into the test tube 1, test tube 2 and test tube 4 into 3.
4. Plug all the tubes with corks and seal them with wax.
5. Wrap tube 2 with aluminum foil or black paper from all side so that light cannot penetrate into this test tube. Similarly wrap test tube 3 with tissue paper.
6. Place all tubes on stand and put the stand in well lighted place.

7. Note and record your observations given below in the table by tick marking the right in the following table:

	Test Tube 1	Test Tube 2	Test Tube 3	Test Tube 4
Light on	Yes/No	Yes/No	Yes/No	Yes/No
Leaf present	Yes/No	Yes/No	Yes/No	Yes/No
Foil on Tube	Yes/No	Yes/No	Yes/No	Yes/No
Indicator color after an hour	Yellow/ Orange/ Red/ Magenta/ Purple	Yellow/ Orange/ Red/ Magenta/ Purple	Yellow/ Orange/ Red/ Magenta/ Purple	Yellow/ Orange/ Red/ Magenta/ Purple
Carbon dioxide concentration	Highest/ High/ Normal/ low/ lowest	Highest/ High/ Normal/ low/ lowest	Highest/ High/ Normal/ low/ lowest	Highest/ High/ Normal/ low/ lowest
Respiration	Yes/No	Yes/No	Yes/No	Yes/No
Photosynthesis	Yes/No	Yes/No	Yes/No	Yes/No

Critical Thinking:

Q1. Is there any change of coloration of Hydrogen bicarbonate indicator?

Q2. What account for these changes?

1.3 GASEOUS EXCHANGE IN ANIMALS

Like plants, animals do exchange gases with their environment for the respiration process. In order to obtain energy from food, they take in oxygen and give out carbon dioxide. So the process of gaseous exchange is ultimately linked with the respiration.

The respiratory medium for aquatic animals is water whereas for terrestrial animals is air. The amount of molecular oxygen present in air is about 21% while in water it is about 5%. In order to exchange gases, animals have a respiratory surface. In unicellular organisms like Protozoa, the plasma membrane serves as the respiratory surface. In multicellular animals, their body surface or some internal surface could serve as the respiratory surface.

Properties of Respiratory surface:

1) Thin, 2) Wet, 3) Permeable, 4) large in relation to the volume of the body.

Proportion of Respiratory surface:

It must be sufficiently large enough to exchange gases for all the cells of the body. For example, the total surface area of the respiratory surface in human is about 20 times to the size of the body.

Respiratory Surfaces
Large surface area:

☞ An increase surface for gaseous exchange allows a faster rate of diffusion to supply oxygen and remove carbon dioxide. It is also necessary to compensate for the small surface area to volume ration of the animal body.

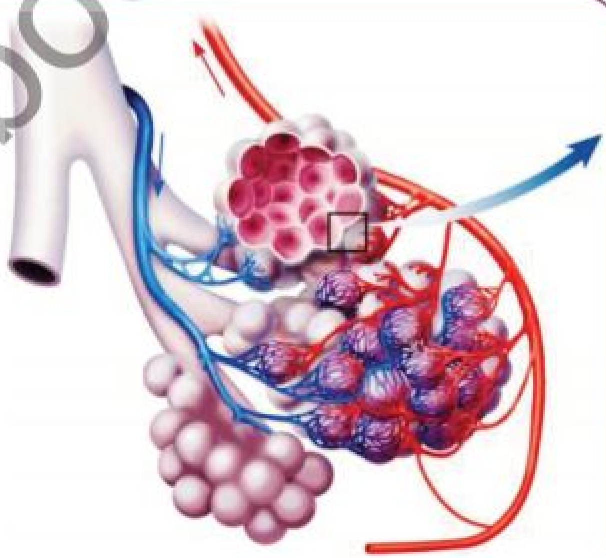


Fig. 1.2 Respiratory Surfaces

1.4 GASEOUS EXCHANGE IN HUMAN

In human, the process of respiration involves breathing, gaseous exchange and cellular respiration. Like other terrestrial vertebrates, our respiratory surface is located inside the body in the form of alveoli contained in paired organs, **lungs**.

1.4.1 Human Respiratory System:

Our respiratory system consists of paired lungs located inside the thoracic (chest) cavity and Air passage ways.

LUNGS:

Each lung is soft, spongy and pinkish in appearance. It is wrapped in two **Pleural membranes**. The space between pleural membranes is filled with fluid that acts like a lubricant. This makes the breathing movements easier. Lungs are enclosed in a bony cage made up of a flat sternum in front, 12 pairs of ribs from front to back where vertebral column is present. Ribs are attached with intercostals muscles. In the lower part of thorax, lies a sheet of muscles called **Diaphragm** which separates it from abdominal cavity. Each lung is made up of millions of **alveoli**.

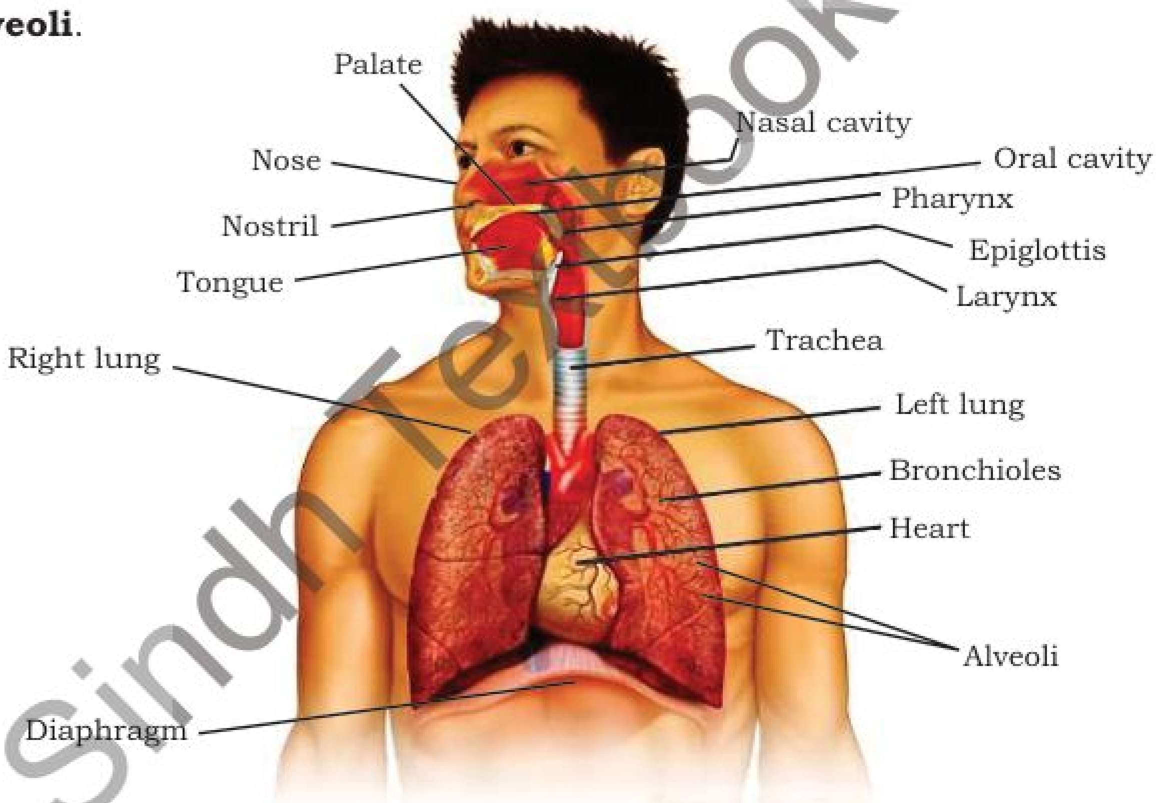


Fig. 1.3 Human respiratory system

Each alveolus is the respiratory surface. It's a pouch like microscopic structure made up of only one layer of cells. It is enclosed by a dense capillary network. In each alveolus, exchange of gases takes place between air and blood. The Air passage ways consists of Nostrils, Trachea, Bronchi and Bronchioles. Air from outside enters into the nasal sacs through external nostrils. This entire passage through which air passes is lined by mucous secreting ciliated cells. The internal surface has rich blood capillaries which turn the incoming air slightly hot. The hairs in nasal sacs as well as ciliated epithelial lining and mucous keep the air clean by trapping and removing dust and germs. This ensures clean air to approach the respiratory surface.

Trachea:

The internal opening of nasal sacs opens into a long tube, **Trachea**. In its beginning, there is a box like **larynx** or sound box containing vocal cords to produce sound. The opening of larynx is known as **glottis** which has a lid like cover, **epiglottis**. During swallowing food or drink, the epiglottis closes the glottis to prevent any food to enter into the trachea. Trachea has C-shaped cartilaginous rings which prevent it from collapsing.

Bronchi:

Trachea in the center of the thorax bifurcates into two smaller ducts or Bronchi. Each bronchus do have C-shaped cartilaginous rings. Bronchus of each side enters into the respective lung. As soon as it enters into the lung, it breaks up into many smaller ducts or Bronchioles.

Bronchiole:

Each bronchiole is very thin tube that opens into **air sacs** or **alveoli**.

1.4.2 Process of Breathing:

Since the respiratory surfaces are located deep inside the body in the lungs. So in order to perform exchange of gases, the air must first be brought into the lungs from the atmosphere. It is achieved through the process of breathing or Ventilation. The process of breathing consists of two phases, viz., Inspiration and expiration.

LUNG CAPACITY: Like a balloon, lungs can be filled with a maximum amount of 5 liters of air. Surprisingly, we use normally about ½ liter of the air coming into the lungs.

Inspiration:

It is the process through which atmospheric air is directed through the air passage ways up to the alveoli in the lungs. It involves contraction of intercostals muscles and diaphragm. As a result, the volume of thorax (chest) is increased thereby decreasing the pressure of air in lungs. So the external air rushes inside from high pressure to low pressure. The lungs get expanded in this way.

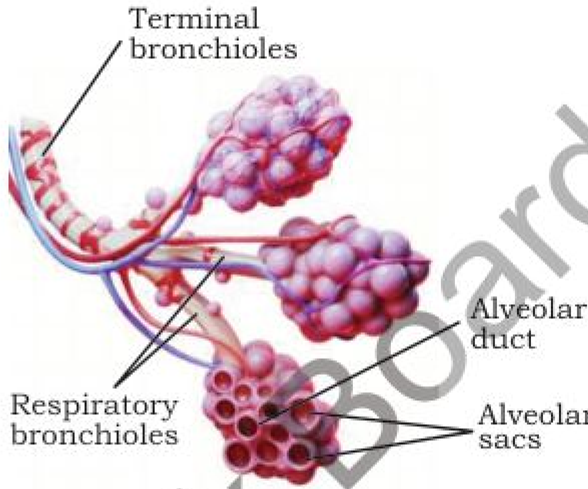


Fig. 1.4 Inspiration

Expiration:

It is just reverse of inspiration. During this process the air moves out from the lungs. Both, intercostals muscles and diaphragm are relaxed. This moves the ribs inside and diaphragm becomes flat. Both the activities depress the chest inside. The volume of thoracic cavity (chest) is decreased causing an increase upon the pressure on lungs. This forces the air present in lungs to outside through the body.

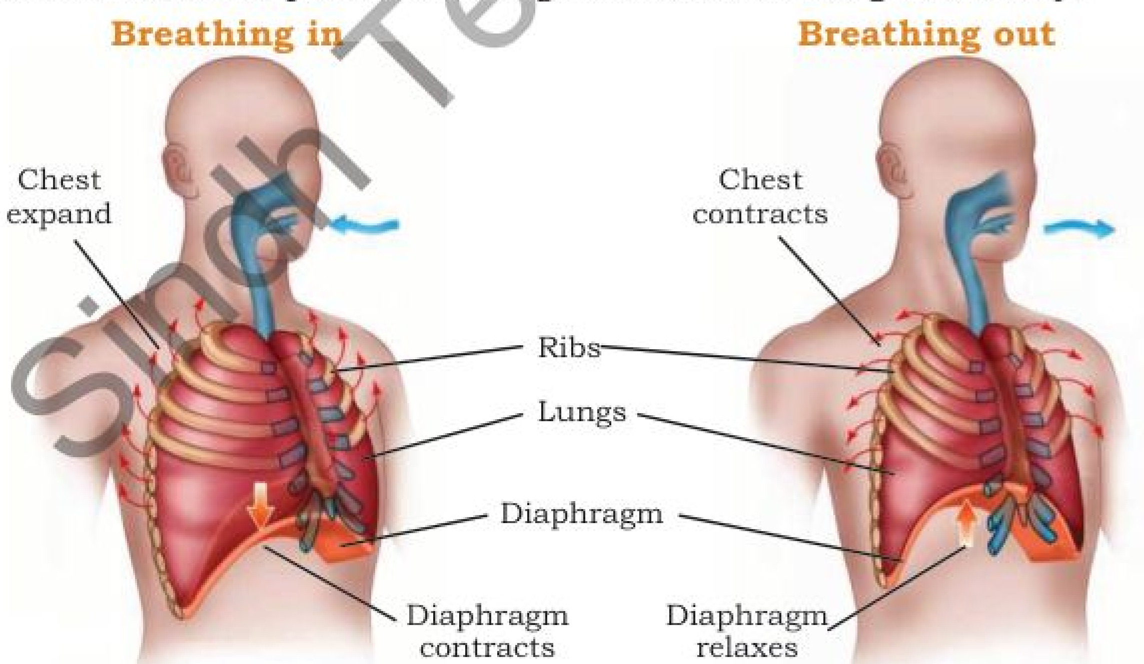


Fig. 1.5 Expiration

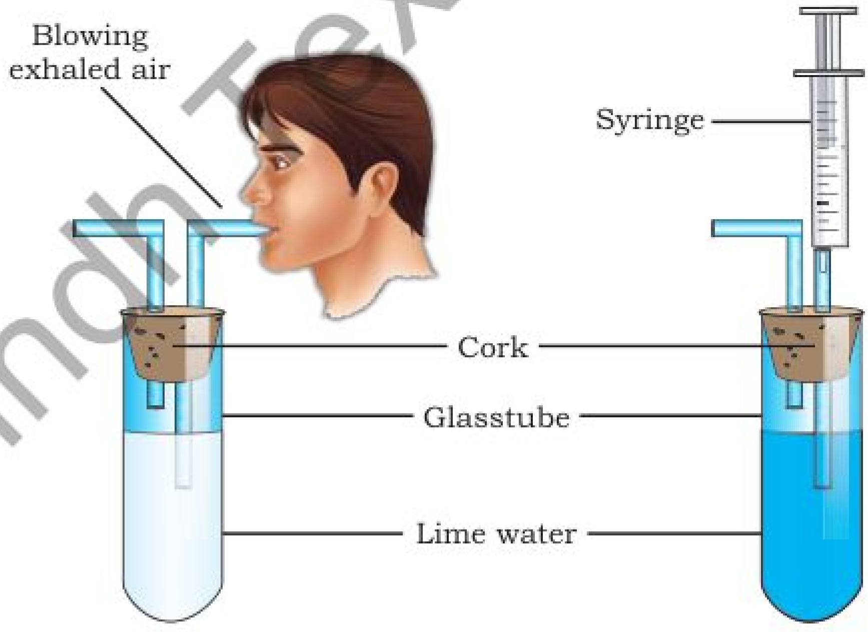
1.5 GASEOUS EXCHANGE IN ALVEOLI:

The gaseous exchange takes place at the level of alveoli. Oxygen brought in by air is taken up by the hemoglobin of RBCs of blood and vice versa the carbon dioxide brought by the blood is given out to the air present in alveoli. This gaseous exchange involves diffusion which becomes possible at this level because both alveolus and blood capillaries, are only one cell layered in thickness.

10.5.1 Composition of Inspired and Expired Air:

Components (%)	Inspired air (%)	Expired air (%)
Oxygen	About 21	About 16
Carbon dioxide	About 0.03	About 4
Nitrogen	About 79	About 79
Water Vapour	Variable	saturated
Temperature	Atmospheric temperature	37 degree celsius

Table 10.1 Composition if Inspired and Expired air



(A)
Air being into lime water

(B)
Air being passed into lime water with a syringe

Fig. 1.6 Demonstration of release carbon dioxide during expiration

Carbon dioxide if passed through limewater turns it milky. This is evident through experiment (A) when one exhales air through the lime-water containing apparatus, the lime water turns milky. In (B) if atmospheric air is passed through the lime-water, the later remains unchanged.

10.5.2 Rate of Breathing at Rest and During Exercise:

Breathing is largely an involuntary process. It is regulated by hypothalamus of our brain. The rate of breathing changes automatically according to the changes in internal or external conditions. For instance, if a person is doing exercise, its rate of breathing would increase because of increased consumption of oxygen by his muscles. Thus gradually increase in concentration of carbon dioxide in his blood will cause an increase his breathing rate. If the exercise condition persists, the muscle cells will start breaking down Glucose without oxygen. It is termed as **"anaerobic respiration"**. As a result of this, lactic acid is formed in the muscles rather than carbon dioxide. It causes pain and cramp in normal muscles. The breaking down of lactic acid requires additional amount of oxygen which is termed as "oxygen debt". The extra amount of oxygen is obtained through deep breathes.

Artificial Ventilator:

A machine that works like lungs when patient's natural breathing becomes difficult. Through this machine, the oxygen rich air is directly supplied to the trachea through a tube inserted the mouth upto the wind pipe.

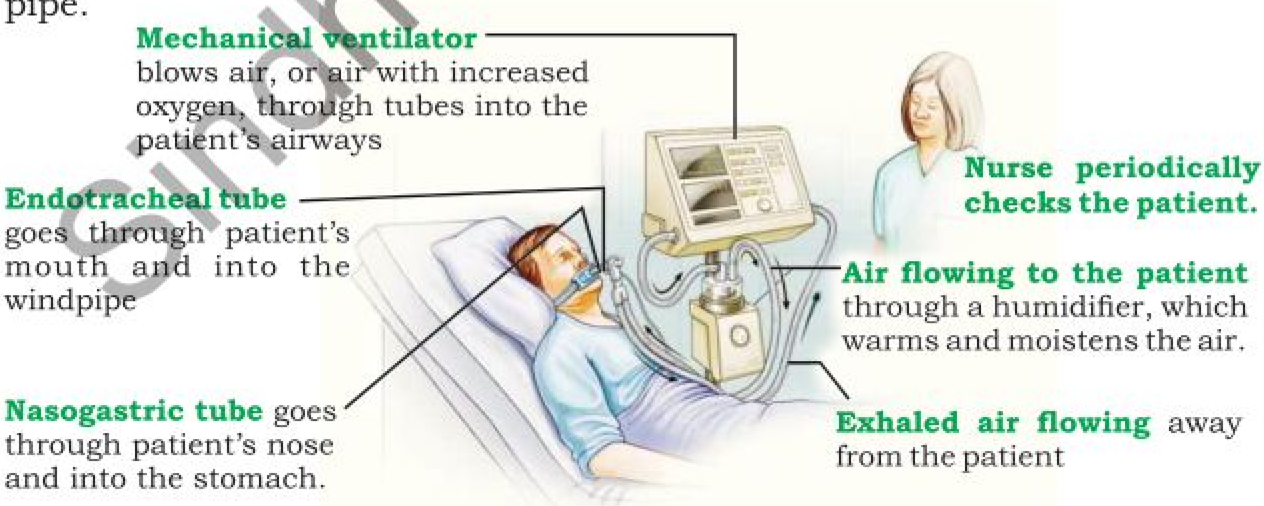


Fig. 1.7 Artificial ventilator

1.6 RESPIRATORY DISORDERS:

Bronchitis:

The inflammation of the air passage ways is termed as **Bronchitis**. It is caused either by smoking or by some bacteria. It is characterized by cough, increased mucous secretion, shortness of breath and low fever.

Emphysema:

It is related to the progressive destruction to the alveoli due to long term exposure usually to the industrial pollutants. It is characterized by laborious breathing. It causes cough with phlegm production.

Pneumonia:

It is an infectious disease usually caused by special bacteria, viruses or fungi. In pneumonia, the alveoli are infected so they may be filled with fluid or pus. The breathing becomes difficult. The patient suffers from fever, cough, chill and chest pain.

Asthma:

It is an inflammatory condition of air-ways of lungs. It is characterized by shortness of breath, chest pain, fever, wheezing sound during expiration and cough. Asthma is actually an allergic response to pollens, dust, smoke, fur, feathers and number of other substances. It may obstruct the air-ways making it difficult to breath for its patient.

Lung Cancer:

Lung cancer is usually associated with smoking. Due to smoke or air pollution, abnormal cells appear in lungs which may spread to other tissues. The major signs and symptoms are cough with blood, shortness of breath, repeated lung infections, weight loss, bone ache, hoarseness, weakness, fatigue, etc.

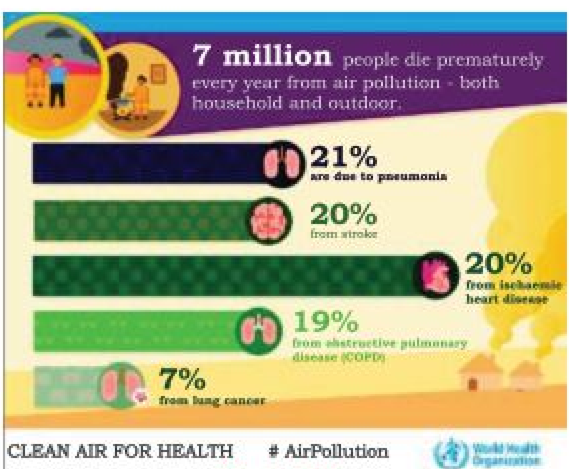


Fig. 1.8
Deaths linked to air pollution

SUMMARY

- Respiration and photosynthesis require exchange of gases.
- Respiration takes place in all living organisms.
- Photosynthesis occurs in green parts of plants.
- During respiration, oxygen is used and carbon dioxide is given out while in photosynthesis, carbon dioxide is used and oxygen is given out.
- In terrestrial plants, most of the exchange of gases occurs through minute openings, stomata.
- The animals use either their body surface, or some internal surface for the exchange of gases.
- Respiratory surface must be thin, wet, permeable and large in relation to the volume of organism.
- The respiratory surface of man is alveoli present in lungs.
- Both lungs have millions of alveoli.
- Air passage ways lead the atmospheric air to alveoli.
- Air pollution causes number of respiratory problems.
- Clean air is essential for better respiratory health.

EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

i) The biological functions which perform gaseous exchange:
(a) Photosynthesis (b) Respiration
(c) Both a & b (d) Growth

ii) Plants do exchange of gases through:
(a) Roots (b) Stomata
(c) Stem (d) All of these

iii) Each stoma is formed by:
(a) one guard cell (b) two guards cells
(c) three guard cells (d) four guard cells

iv) Respiratory surface possesses following property:
(a) thin and wet (b) permeable
(c) very large (d) all of these

v) Inspiration involves:
(a) Contraction of intercostals muscles
(b) contraction of diaphragm
(c) Inward movement of ribs
(d) Both a & b

vi) Larynx is located on:
(a) Lungs (b) Trachea
(c) Bronchus (d) Bronchiole

vii) The respiratory surface of human is:
(a) Nostril (b) Bronchiole
(c) Alveoli (d) Trachea

viii) Increase in rate of breathing is due to the following:
(a) increase CO_2 in blood (b) Increase O_2 in blood
(c) decrease CO_2 in blood (d) decrease O_2 in blood

ix) Which of the following disorder is associated with degeneration of alveoli?

(a) Bronchitis (b) Lung cancer
(c) Asthma (d) Emphysema

x) Which of the following disorder is associated with inflammation of air passage ways?

(a) Bronchitis (b) Lung cancer
(c) Asthma (d) Emphysema

B. SHORT QUESTIONS:

i) Why the stomata generally open during day-time?

ii) Which parts of the plant intake CO_2 and give out O_2, take in O_2 oxygen and give out CO_2 during day-time?

iii) Why do we have to breathe through nostrils rather than oral cavity?

iv) Differentiate between breathing, gaseous exchange and respiration.

v) Why do we deep breath during or immediately after exercise?

vi) What is "oxygen debt"?

vii) Distinguish between inspiration and expiration.

viii) What is lung cancer?

ix) How the asthma is characterized?

x) Name five animals which use their body surface for gaseous exchange.

C. EXTENSIVE RESPONSE QUESTIONS:

i) What measures would you take to avoid respiratory disorders?

ii) Discuss human respiratory system with the help of suitable illustrations.

iii) Prove with the help of experiment that CO_2 is released during respiration.

iv) Explain the process of ventilation in man.

v) Why smoking is dangerous? How it is related with respiratory disorders?

Chapter 2

HOMEOSTASIS

Major Concept

In this Unit you will learn:

- Introduction
- Homeostasis in Plants
- Homeostasis in Man
- Homeostasis in Animals
- Urinary system of Man
  - Structure and Functioning of Human Kidney
  - Structure of Kidney
  - Structure of Nephron
  - Functioning of Nephron
- Disorders of Human Excretory System
- Kidney Stones and Treatment